

The land feels it *first*.

NABD — early warning that reaches the person, in their language, even off the grid

- نبض · "pulse" in Arabic
- Built in the UAE for sites where help is hours away

نبض

The warning existed. It didn't reach the *person*.

- 16 April 2024: 254.8 mm of rain fell in under 24 h at Khatm Al Shakla, the heaviest since records began in 1949 (NCM). At least four people died in the UAE, and Dubai airport cancelled 1,244 flights over two days. Warnings were issued in advance.
- Qatar 2009–2017: cardiovascular deaths among young Nepali workers rose with summer heat. As many as 200 of 571 could have been prevented with effective heat protection (Pradhan et al., Cardiology, 2019).
- Across the Gulf, as many as 10,000 South and Southeast Asian migrant workers die each year, and more than half of those deaths are certified with no underlying cause (Vital Signs Partnership, 2022).



16 APRIL 2024 · UAE

254.8 mm

of rain in under 24 h, the most since 1949

Warnings went out in advance. People still drove into wadis; crews were still on site.



QATAR · 2009–2017

200 of 571

young Nepali workers' heart deaths could have been prevented

Deaths rose with summer heat. Effective heat protection was the missing piece.



ACROSS THE GULF · EVERY YEAR

“Natural causes”

on more than half of up to 10,000 deaths

Migrant workers' deaths are certified with no underlying cause, so heat never shows up.

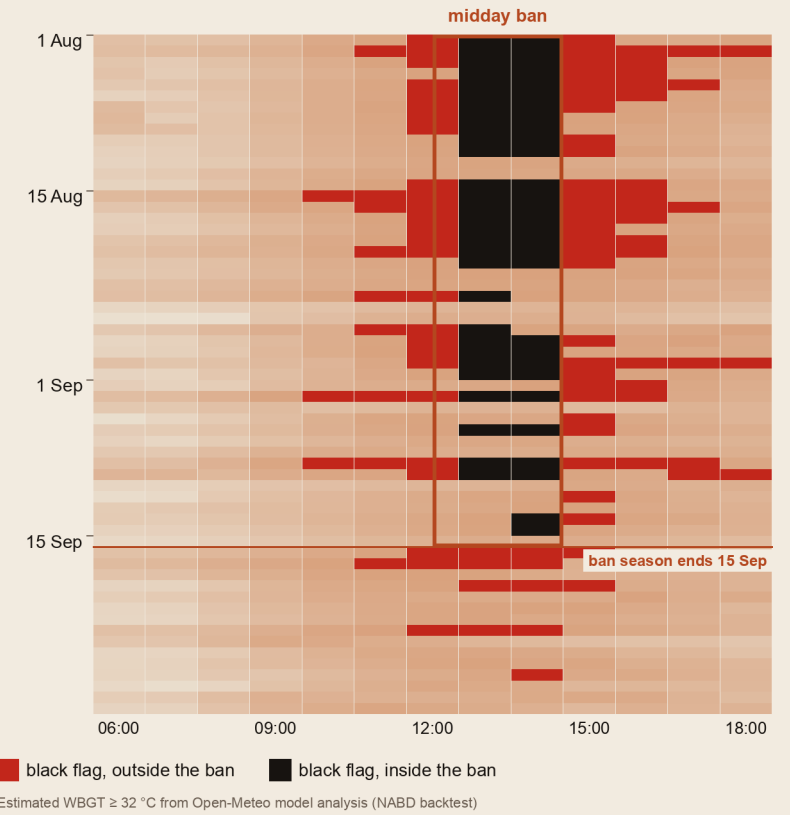
Rules follow the *calendar*. Danger doesn't.

- Calendar vs physics: the UAE midday ban covers 12:30–15:00, 15 June –15 September. In August 2025 at Dubai South, 65 of 113 black-flag heat hours fell outside it (our backtest)
- Region vs site: forecasts speak about "the eastern region"; a supervisor needs "stop the crew at block C at 11:00"
- Language: alerts go out in Arabic and English; crews read Hindi, Malayalam, Tagalog, Urdu, Bengali
- Connectivity: offshore rigs, Hatta wadis and desert farms lose signal exactly when it matters

100 of 158

black-flag work hours fell outside the midday ban

Dubai South · 1 Aug – 30 Sep 2025 · 06:00–18:00



The Gulf is the planet's *preview*

83.6%

of workers in the Arab States are exposed to excessive heat, against a 71% global average (ILO, 2024)

18,970

work-related deaths a year worldwide attributed to excessive heat (ILO, 2024)

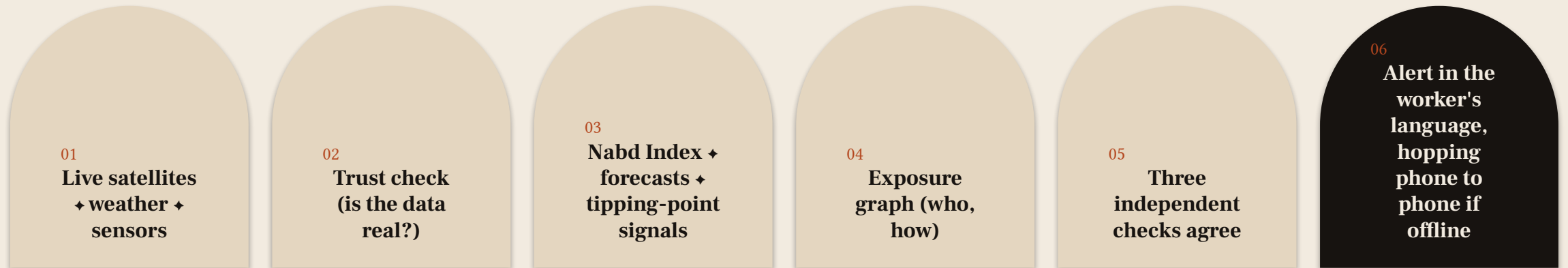
2.2%

of all working hours projected lost to heat stress by 2030, equal to 80 million jobs and US\$2.4 trillion (ILO, 2019)

35 °C

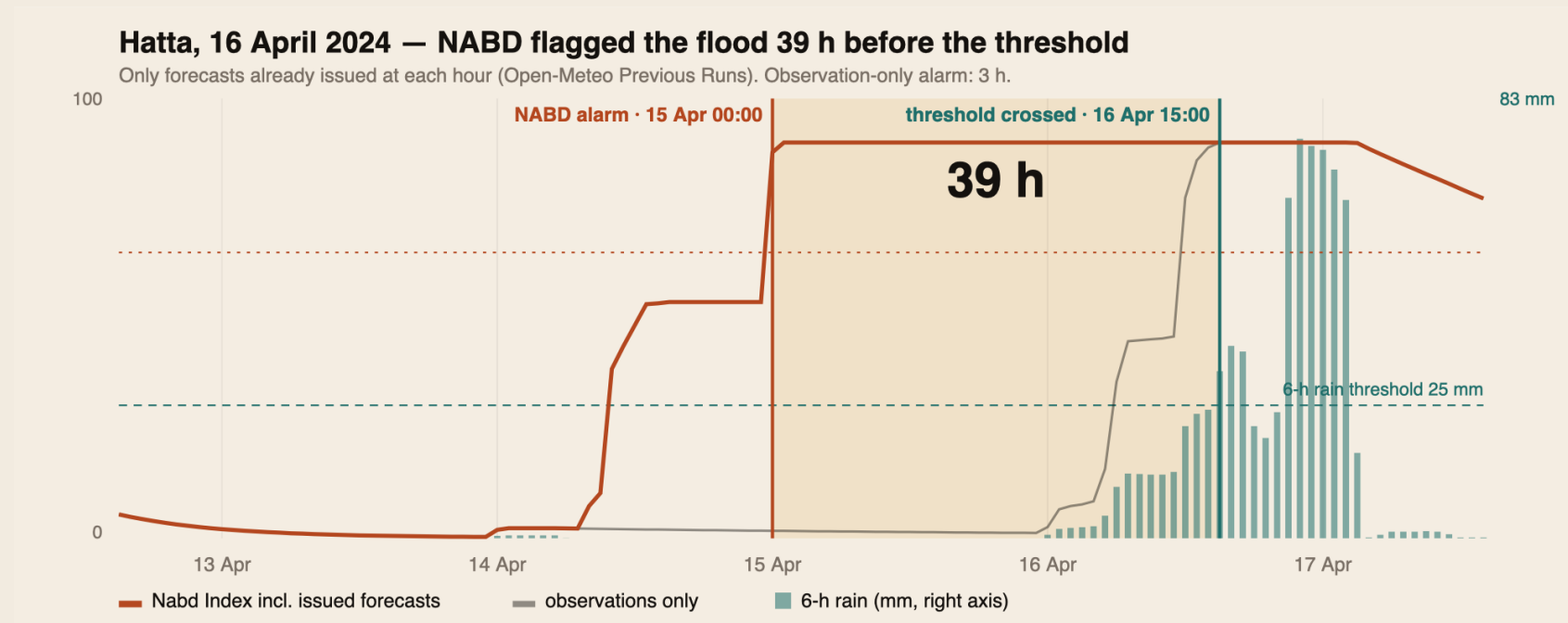
wet-bulb: the survivability limit that Abu Dhabi, Dubai and Doha are projected to exceed by 2071–2100 under high emissions (Pal & Eltahir, 2016)

From the land's signal to the person's phone



- Free, keyless live sources: Open-Meteo, NASA GIBS, NASA EONET, geoBoundaries, WorldPop
- Every decision is computed, explained and written to a hash-chained audit log; Gemini only writes the optional briefing

Hatta, April 2024: flagged *39 hours* before the flood threshold

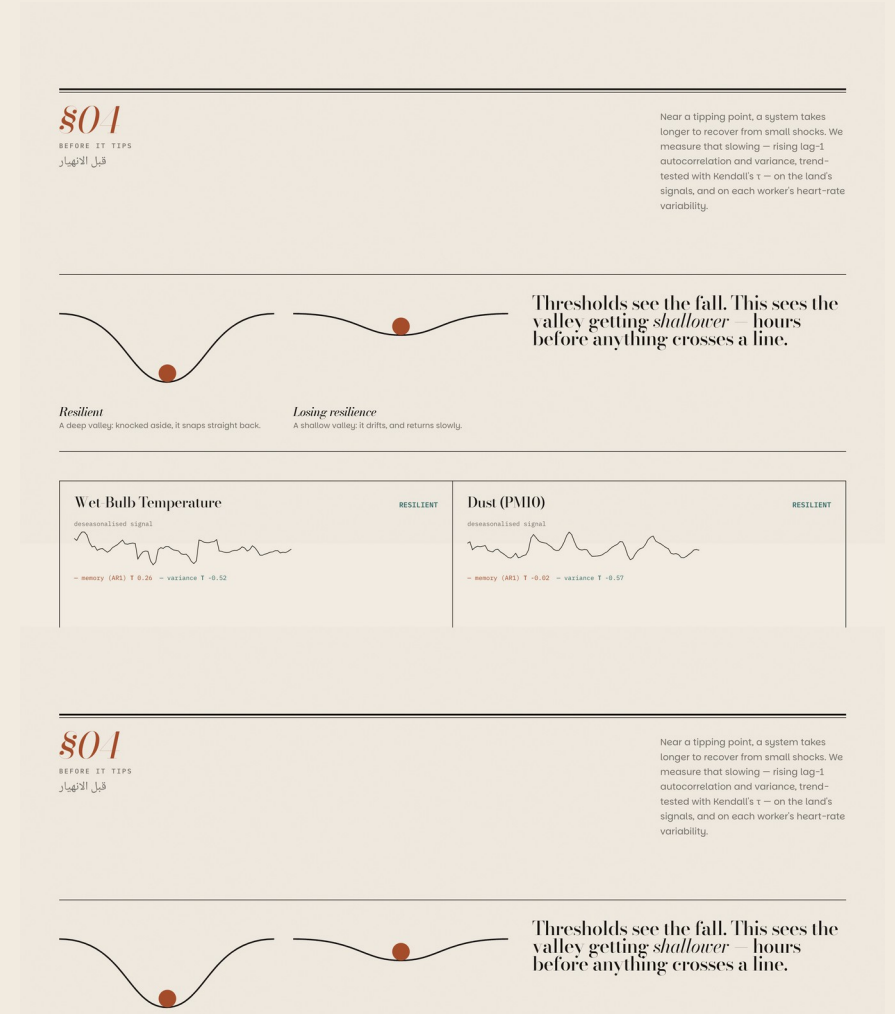


- Replayed with only the forecasts already issued at each hour (Open-Meteo Previous Runs); a test proves no future data leaks in
- Heat, Dubai South: 25/25 black-flag days flagged by 05:00 in Aug 2025 (median 7 h ahead, false-alarm ratio 0.19); 14/14 in Sep 2025 (false-alarm ratio 0.53)

Honest: the lead comes from public forecasts plus the index's memory; flood false-alarm periods were 7 of 9; reference is model analysis, not gauges

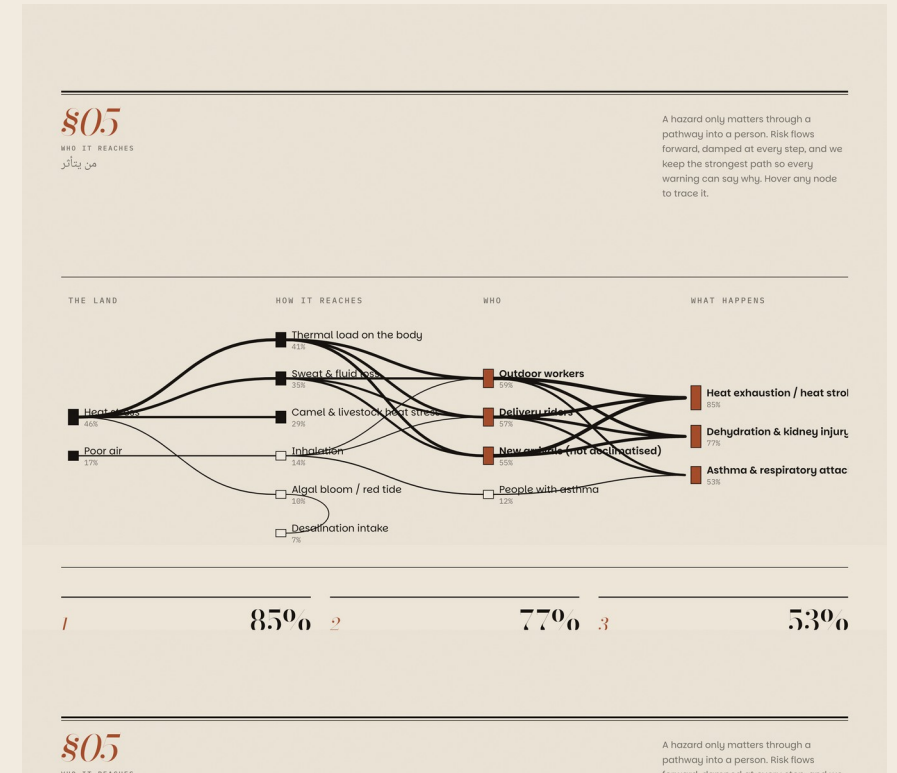
A number with *memory*, plus a check for lost resilience

- Nabd Index (0–100): each hazard fades at its own pace: heat in hours, dust in a day, a warm sea in three days
- Combined with the forecast, it gives the lead time on the validation slide
- Critical-slowness statistics (rising autocorrelation and variance, daily cycle removed, tested against random noise) add one level of caution, never an alarm on their own



Not "it's hot." *Who* gets hurt, and *how*.

- Hazard → pathway → cohort → outcome, with the strongest path kept so every warning explains itself
- Gulf-specific: warm sea → algal bloom → desalination intake → drinking water. The 2008–09 red tide forced desalination plants in the UAE and Oman to shut or cut output (Richlen et al., 2010)
- Weights are tunable priors, labelled as such, for each site to calibrate



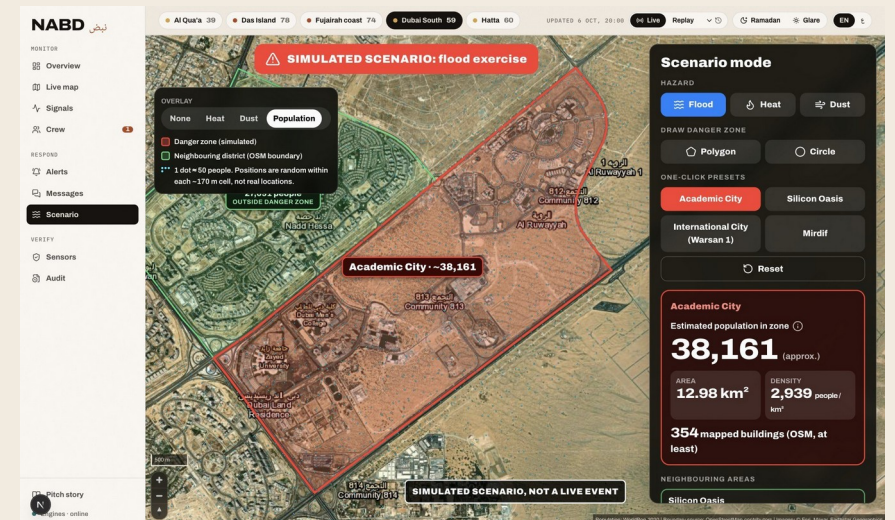
Day 3 in Dubai is not *year 9*.

- Moran Physiological Strain Index per worker from heart rate and skin temperature; the limit moves with acclimatisation, age and Ramadan fasting
- Built for WHOOP bands and Apple Watch (WHOOP API, Apple HealthKit); simulated from real site WBGT today
- A bounded Bayesian belief learns from supervisor feedback, so it can never switch protection off
- Symptoms typed in Hinglish, Arabizi, Malayalam or Tagalog raise that worker's strain, and the reply comes back in their language



Draw the flood zone. See *who is inside*.

- Real district boundaries from OpenStreetMap: Academic City (12.98 km²) gives ~38,200 modelled residents and at least 354 mapped buildings
- People drawn as dots: 1 dot ≈ 50 people, scattered randomly inside each ~170 m WorldPop cell
- Exposure watch: live weather sets a safe time in the sun; anyone past it gets a WhatsApp alert in their own language (demo)
- Labelled every time: simulated people and zone, real conditions, works offline from saved files



Works when the network *doesn't*, and when someone lies

- A 21-byte signed alert fits one Bluetooth LE advertisement and hops phone to phone, with store-and-forward
- Three independent checks (land, body, data) must agree; untrusted data means hold, never "all clear"
- Signed sensors with physics and satellite cross-checks, attacked by our own red team



Compliance, fewer incidents, *more safe working hours*

AED

50,000

maximum midday ban fine (AED 5,000 per worker, MoHRE 2025); a single maximum fine covers about 20 months of a site licence

2.2%

of working hours lost to heat by 2030; shifting work by the forecast instead of losing the day protects output (ILO, 2019)

1 log

hash-chained, inspector-ready proof of every warning and action, for HSE audits, insurers and ESG reports

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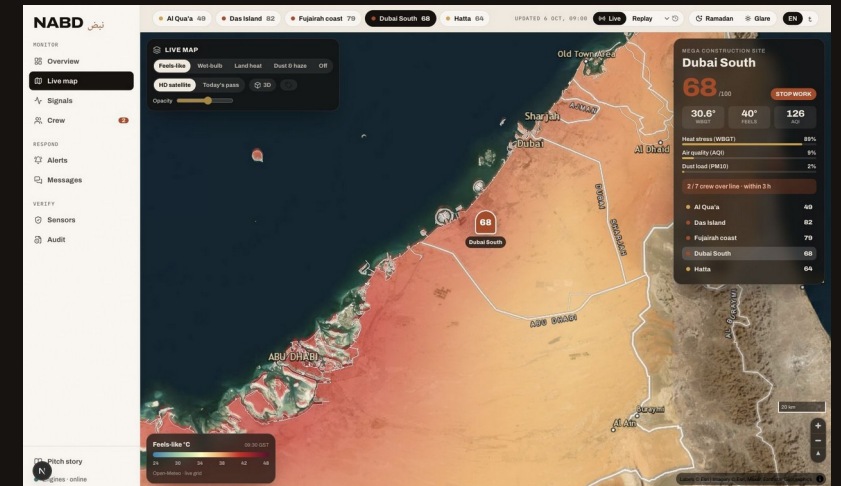
paid APIs: free public data keeps the cost per site low

Who buys first, and *how*

- Contractors and mega-projects: crews of hundreds per site, midday-ban liability
- Oil & gas and utilities: offshore crews, desalination intakes, strict HSE programmes
- Delivery and logistics fleets: riders under open sky all day
- Government: emergency management, municipalities, labour inspection; Scenario mode for flood planning
- Model: per-site subscription plus per-worker seats (pricing hypothesis: ~AED 2,500 per site per month)
- Channel: HSE consultancies and insurers, who already sell safety to the same buyers
- Wedge: start with one contractor's sites for the summer season, prove fewer incidents and zero fines, then expand
- Expansion: the same engine for Saudi giga-projects, Qatar, Oman, and any hot, remote workforce

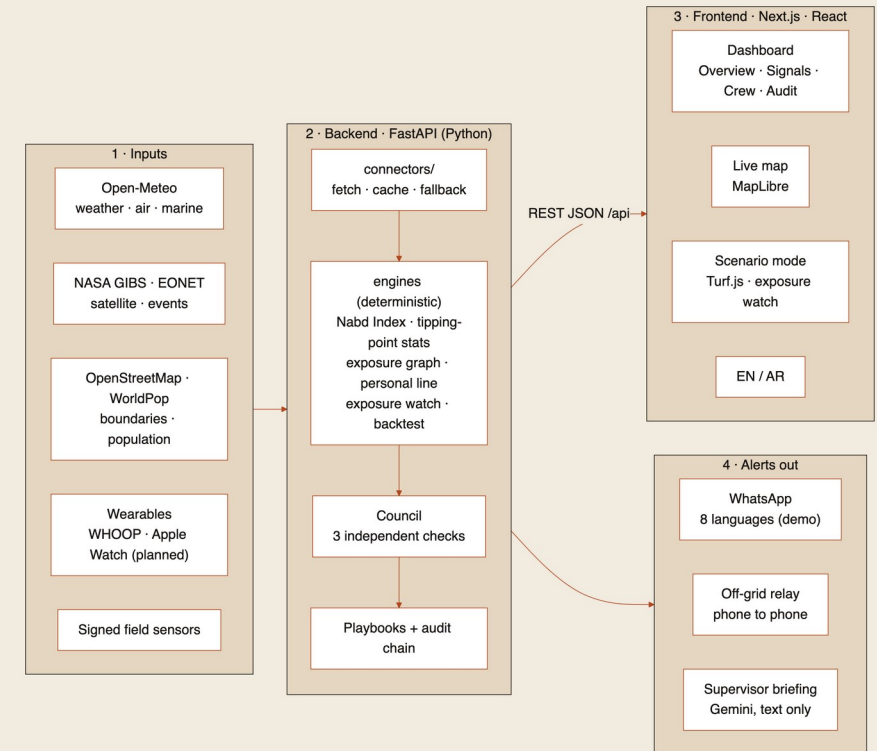
Live data, *simulated scenarios*

- 1. Overview: five real UAE sites, index, verdict, crew over their line
- 2. Live map: HD satellite, live feels-like heat field, NASA land-heat and dust
- 3. Signals: tipping-point statistics, plus the validation backtest
- 4. Scenario: Academic City gives people and buildings; exposure watch drafts a WhatsApp sun alert (demo)
- 5. Alerts: send "Stop work", watch it hop phone to phone in five languages
- 6. Messages: a code-switched symptom message fires the playbook



Free public data in, *alerts* out

- Frontend: Next.js and React, MapLibre maps, Turf.js geometry, English and Arabic
- Backend: Python FastAPI; connectors fetch, cache and fall back; deterministic engines decide
- Inputs: Open-Meteo, NASA, OpenStreetMap, WorldPop, signed sensors; WHOOP and Apple Watch next
- Out: WhatsApp alerts in 8 languages (demo), the off-grid phone relay, and a hash-chained audit log



What we *haven't* proven yet

- Lead time comes from public forecasts plus memory; tipping-point statistics are context
- Heat alarms catch every black-flag day but over-warn: false-alarm ratio 0.19 (Aug) and 0.53 (Sep); flood alarm periods 7 of 9 false
- Wearables are simulated from real site WBGT; WHOOP and Apple Watch not yet connected
- Mesh: iOS limits background Bluetooth; a shared crew key means a key holder could forge alerts
- Exposure-graph weights are tunable priors, not fitted
- Pilot one contractor site with WHOOP bands or Apple Watch (WHOOP API, Apple HealthKit)
- Per-device keys signed by the site office, with revocation
- Satellite chlorophyll and vegetation (NDVI) feeding the ecosystem side; sources already verified
- Calibrate graph weights and thresholds with local incident data

Listen to the land. *Protect the
person.*

github.com/Shreeya1-pixel/Qwen

نبض